# Introduction to Cybersecurity

Network Security




Jasmin Azemović
CISO | Associate Professor | Microsoft MVP

# Summary

- System security
- Network Security
- Network attacks

# System security

- It is much more than:
  - administrator accounts
  - user accounts
  - classical sharing resources
  - It refers to the OS as a whole without the application layer
  - who should "know how to behave" as part of the OS

# OS classification

- 1983 DoD - Trusted computer system evaluation criteria (TCSEC) ", also known as the" Orange Book
- It defines the following classes of operating systems:
  - Class D - Minimum Protection (MS-DOS, Windows 95, 98)
  - Class C1 - Discreet protection (UNIX)
  - Class C2 - Controlled Access Security (UNIX, Linux, NT, 2000, XP ... Win 7, 8, 10)
  - Class B1 - includes elements of confidentiality
  - Class B2 and B3 - Structural and the hierarchical protection from the elements alert
  - Class A1 - Top systems (mainly military custom solutions and real-time systems)

# Network securitity

- What are the problems?
- Network distribution
- Basics protocols are not made with elements of security
- Errors in existing protocols
- DDoS attacks

# Firewalls

- Dedicated components that are located between the Internet / LAN / WAN environments
- Software and hardware
- Complete content packet is subject to passing "through" firewalls
- Applying local security policies
- The base feature is the drop or pass network TCP / IP packets
- Check the destination and source address
- Checking the package contents
- Logging / auditing / alarm
- VPN tunneling

# Firewall limitations

- Little protection for internal traffic
- Aggravation of business processes
- Implementation of new services

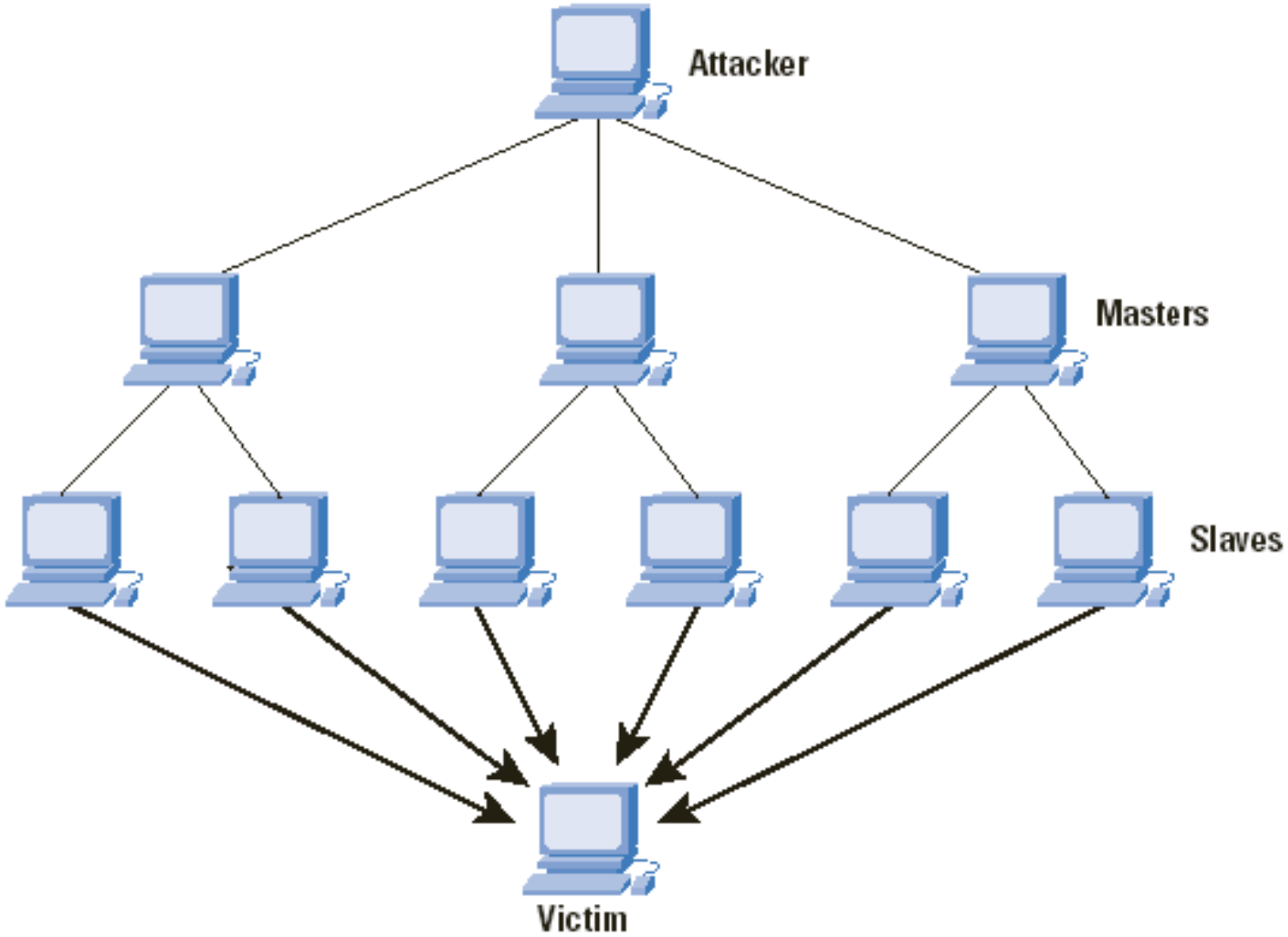
# Why network attacks?

- Motives:
- Denial of network services and service
- Concealing the true reason attacks
- Theft
- Espionage
- Political reasons:
- Information warfare

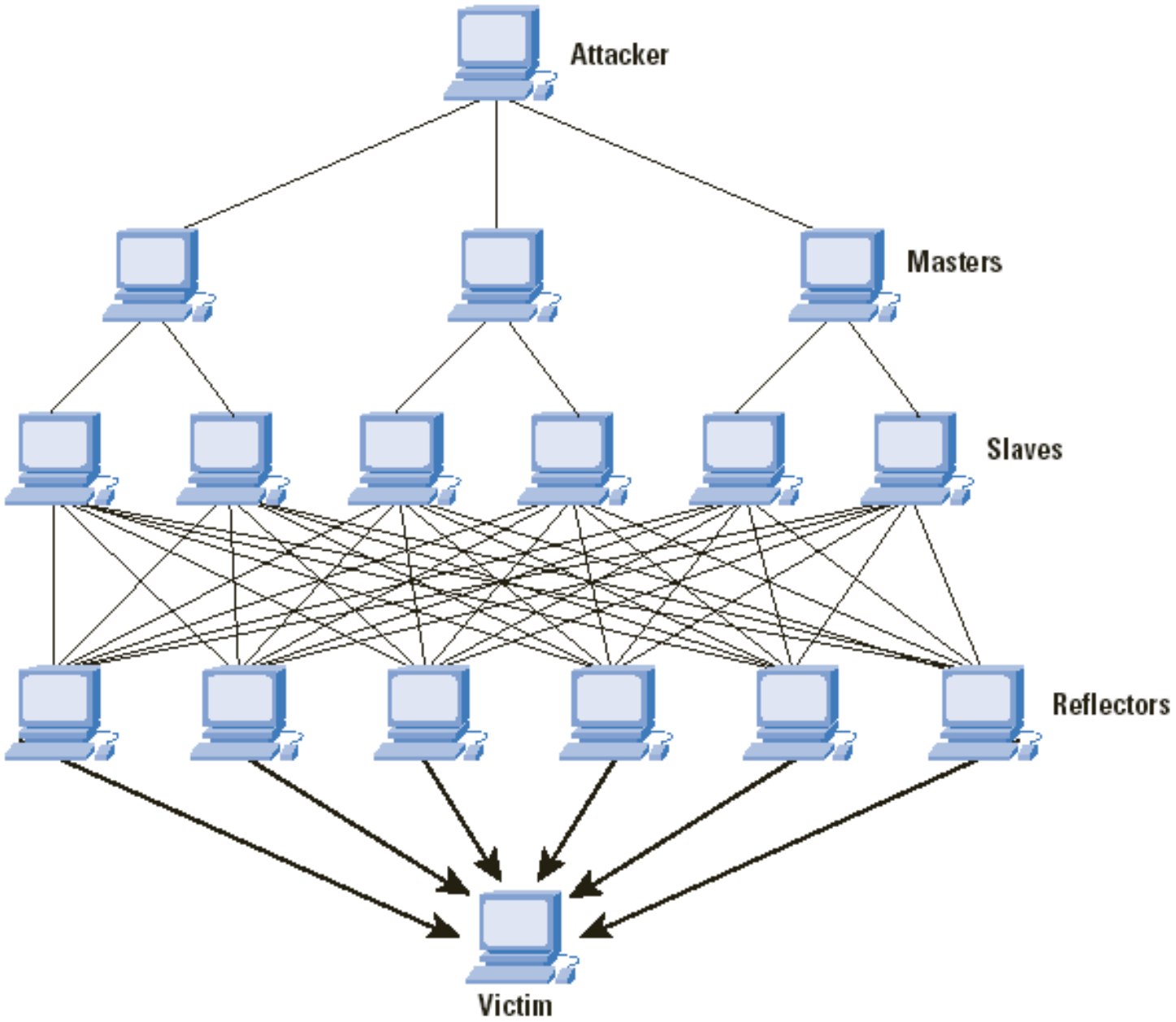
# Who's to blame

- Man
  - psychology
  - naivety
  - ignorance
- Protocols
  - Poorly designed and implemented in such network protocols
- Technology
  - Gaps in the software and hardware components
  - TCB is poorly designed and implemented

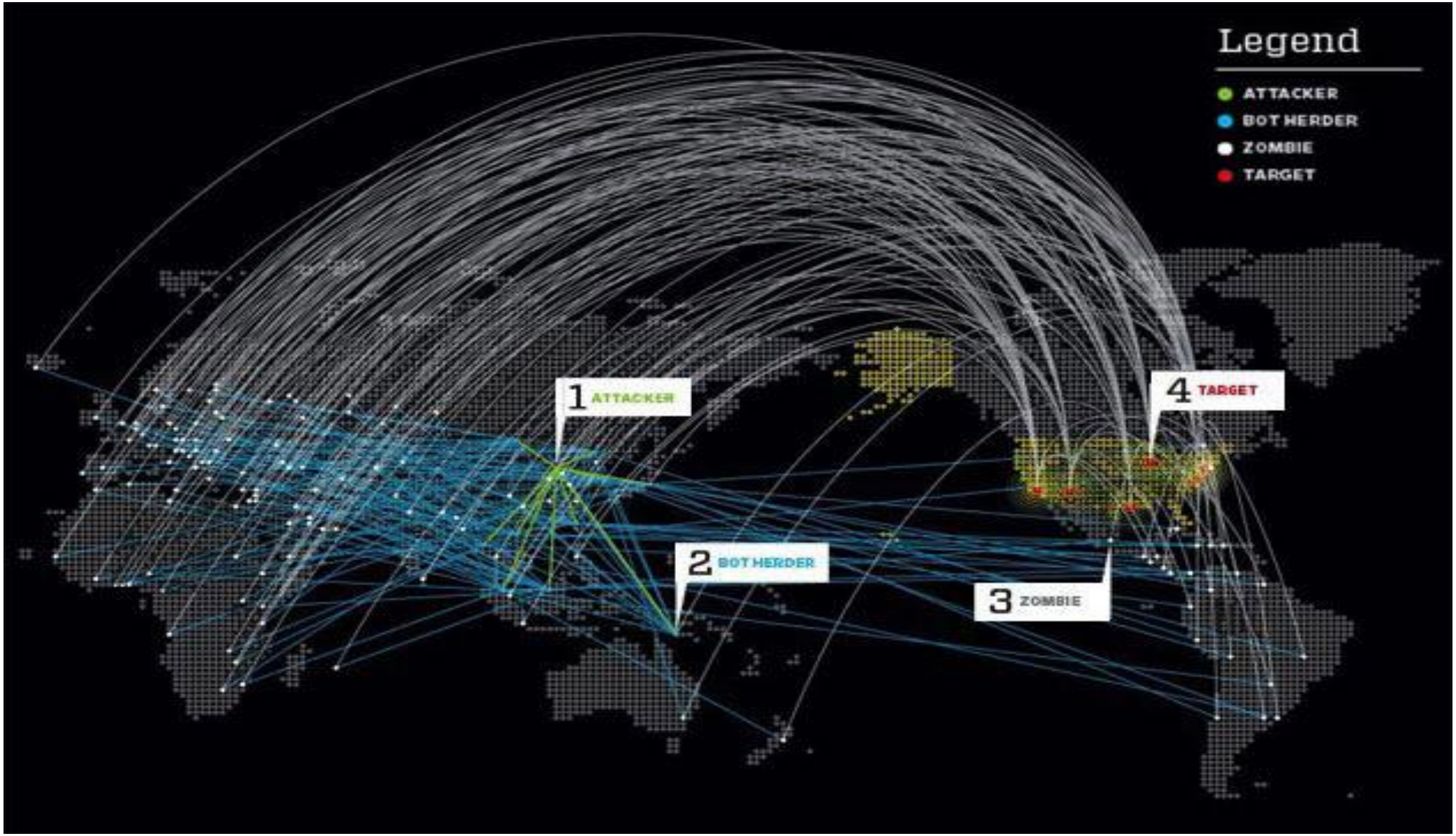
# DoS/DDoS



# DoS/DDoS

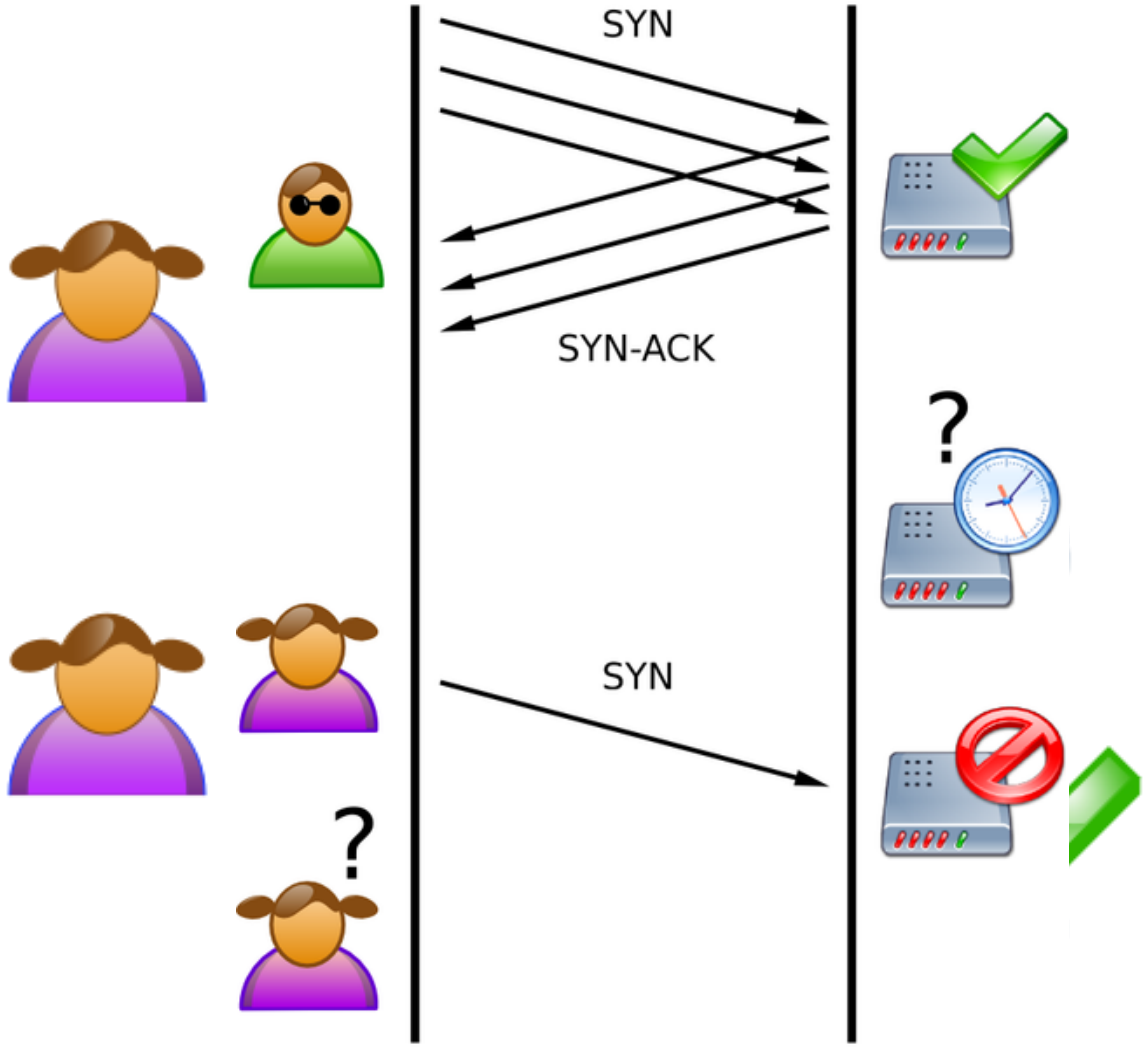


# DoS/DDoS


Legend
ATTACKER
BOT HERDER
ZOMBIE
TARGET
1 ATTACKER
2 BOT HERDER
3 ZOMBIE
4 TARGET

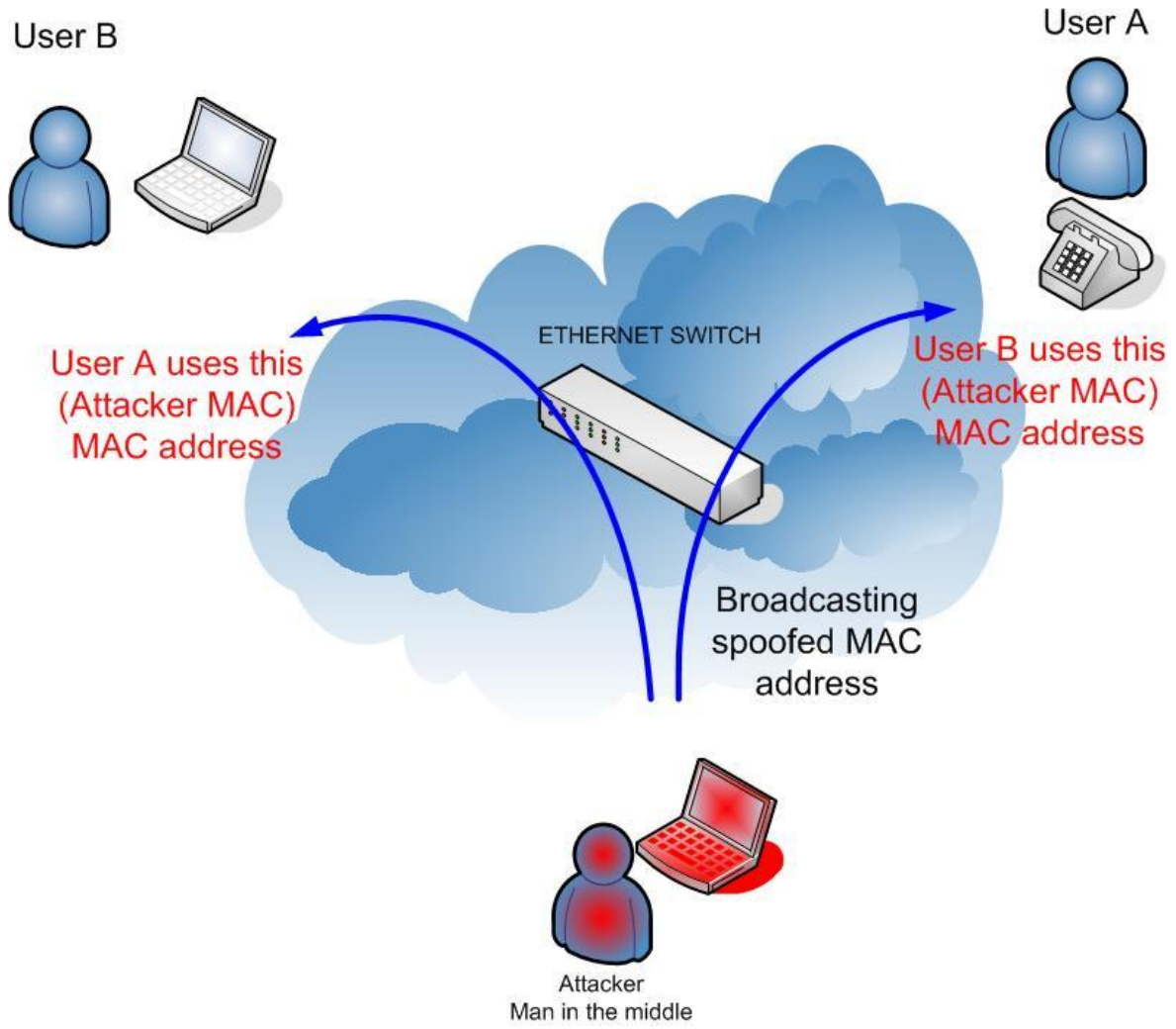
# SYN Flooding

- Defend:
  - SYN Cookies
  - Timeframe limit



# Arp Poisoning

- Defend:
  - IP2MAC maping
  - Port locking
  - Detection software



# Conclusion

- System management
- Filtering network traffic
- Monitoring
- Intrusion detection
- encryption

